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SYSTEMIC REVIEW

Pharmacogenomic studies of fertility outcomes in pediatric cancer survivors – A systematic review

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Abstract

For the same age, sex, and dosage, there can be significant variation in fertility outcomes in childhood cancer survivors. Genetics may explain this variation. This study aims to: (i) review the genetic contributions to infertility, (ii) search for pharmacogenomic studies looking at interactions of cancer treatment, genetic predisposition and fertility-related outcomes. Systematic searches in MEDLINE Ovid, Embase Classic+Embase, and PubMed were conducted using the following selection criteria: (i) pediatric, adolescent, and young adult cancer survivors, below 25 years old at the time of diagnosis, (ii) fertility outcome measures after cancer therapy, (iii) genetic considerations. Studies were excluded if they were (i) conducted in animal models, (ii) were not published in English, (iii) editorial letters, (iv) theses. Articles were screened in Covidence by at least two independent reviewers, followed by data extraction and a risk of bias assessment using the Quality in Prognostic Studies tool. Eight articles were reviewed with a total of 29 genes. Outcome measures included sperm concentration, azoospermia, AMH levels, assessment of premature menopause, ever being pregnant or siring a pregnancy. Three studies included replication cohorts, which attempted replication of SNP findings for NPY2R, BRSK1, FANCI, CYP2C19, CYP3A4, and CYP2B6. Six studies were rated with a high risk of bias. Differing methods may explain a lack

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of replication, and small cohorts may have contributed to few significant findings. Larger, prospective longitudinal studies with an unbiased genome-wide focus will be important to replicate significant results, which can be applied clinically.

INTRODUCTION

The global incidence of childhood cancer is 1–2 cases per 10,000 children under 15 years of age.[1] In developed countries, over 80% of children survive their cancer diagnosis[2] and long-term survivorship issues are now of increasing importance. One long-term morbidity of concern to childhood cancer survivors is infertility.[3,4] Fertility is impacted by cancer treatment, including alkylating chemotherapeutics, radiation therapy, and surgery. To prevent the psychological regret and distress of survivors, it is important to find ways to accurately predict infertility risk and seek interventions or methods to mitigate it.

Predicting infertility risk due to cancer treatment is not an exact science. It is known that age, chemotherapy, and radiotherapy dosage, as well as sex, play a role.[5] Even when researchers control for these factors, variations in fertility outcomes remain. For females, some variation may be secondary to the normal variation in ovarian reserve.[6] Individuals with a greater ovarian reserve prior to commencing cancer therapy may retain more follicles post-therapy. Age also contributes to approximately 80% of the variation in primordial follicle numbers and is the key determinant of reduced oocyte quality in women >36 years of age, while other determinants, including genetic and lifestyle factors, are thought to make a 3%–5% contribution to the age of menopause.[7] Similarly, natural variation is seen in male sperm production.[8] This inter-individual variability indicates a potential genetic impact, which may affect fertility outcomes post-cancer therapy.

The genetics of fertility is an already complex picture in the general population, without considering the extra confounders and sub-populations of childhood cancer patients, such as the types of cancers, varied treatments, the pharmacokinetics of metabolizing chemotherapy drugs, and the pharmacogenomics of chemotherapy-induced damage and recovery from cancer treatment. Therefore, we will first briefly review what is known of these confounders before considering the pharmacogenomics of infertility in childhood cancer survivors.

Genetics of infertility

Several congenital syndromes can lead to significant disruption to gonadal development or fertility, such as early depletion of ovarian reserve, which can be further exacerbated by cancer treatment. Turner syndrome is the most well-known, affecting approximately 1 in 1500 newborn females.[9] Down syndrome is a more common syndrome, affecting 1 in 700 births. Fertility issues are rarely reported, yet reduced attempts at parenthood may influence this. Men and women with Down syndrome appear to have normal puberty but have lower sperm count and ovarian reserve, leading to oligospermia or azoospermia and premature ovarian insufficiency (POI), respectively.[10]

Other rare conditions influencing gonadal function and fertility include congenital disorders of glycosylation type 1a (PMM2-CDG), congenital adrenal hyperplasia (CAH), blepharophimosis, ptosis, and epicanthus inversus syndrome (BPES), aromatase deficiency, Perrault syndrome, ataxia-telangiectasia, Fanconi anemia, Nijmegen breakage syndrome (NBS), and premature aging syndromes like Werner syndrome, Bloom syndrome, Hutchinson-Gilford progeria syndrome, Rothmund-Thomson syndrome, and GAPO syndrome.[11,12] Biallelic defects in DNA repair genes MCM8, MCM9, XRCC4, MSH5,[13] and NBN[14] and FANCM[15] are associated with these syndromes as well as non-syndromic POI.

Direct genetic studies of ovarian reserve at birth are rare since most genetic studies are in older cohorts of POI patients using hormone levels as a surrogate marker for ovarian reserve. In a study by Cunniff (1991), the ovaries of 36 female fetuses, with a variety of chromosome abnormalities were analyzed at autopsy.[16] The results showed that many chromosome abnormalities, including those of chromosomes 2, 4, 11, 13, 18, 20, 21, X, can lead to little or no follicles present. Russell (1975) and Hojager (1978) also supported this finding, regarding trisomy 18 and Down syndrome women, respectively, and references therein.[17,18] Given the number of steps involved in ovarian development and function, the diversity of possible genetic causes of low ovarian reserve is not surprising. Disruptions may lead to accelerated apoptosis and early depletion of the primordial follicle pool.[13]

For males, differences in sexual differentiation that can lead to male hypogonadism can result from both endocrine issues, as discussed in the review by Grinspon (2020),[19] and non-endocrine causes, as outlined in the review by Grinspon (2014).[20] Several candidate genes have been proposed for association with azoospermia,[21–24] yet half of male infertility cases are classified as idiopathic.[25] The main contributing factor to male infertility is assumed to be genetic abnormalities, with known genetic

abnormalities responsible for between 15% and 30% of cases, which include more than 70 genetic syndromes, although most of these syndromes are extremely rare.[26] The most common genetic cause is Klinefelter syndrome, followed by Y chromosome microdeletions, as reviewed in several articles.[25–27] However, environmental factors of testicular dysgenesis syndrome were found to have a stronger influence than genetic causes.[28]

Genetics of POI

It has long been recognized that abnormalities in the X chromosome often lead to POI,[29] with notable cases being Turner and Fragile X syndrome. In fact, there appears to be a critical region between Xq13 and Xq27 that greatly influences POI.[16,29–31] Only a few genes on the X chromosome have been linked with ovarian development and POI including BMP15, FMR1, PGRMC1,[13] and POF1B.[32] X chromosome deletions, duplications, inversions, complex rearrangements, X-autosome translocations, and single-gene sequence variants can otherwise lead to accelerated loss of primordial follices and to recurrent fetal losses.[13]

Aside from X chromosome abnormalities, the full picture is still emerging regarding the genetics of POI. There are over 100 genes that can have POI-associated variants, emphasizing both the scale and complexity of this condition. Various candidate genes and chromosomal abnormalities have been analyzed for associations.[11,33–36] BRCA1 and BRCA2 mutations have been associated with diminishing ovarian reserve measured via anti-Mullerian hormone (AMH),[37–41] as well as lower oocyte numbers and low response to ovarian stimulation in fertility preservation attempts.[42] A recent case study of a consanguineous family showed an association of a biallelic BRCA2 variant with POI.[43] Currently, only testing for FMR1 premutations of Fragile X Syndrome[44] and microarray/karyotyping to detect chromosome rearrangements are routinely offered to POI patients.[36] The status of the genetics of POI has been reviewed recently.[13,36,45–47]

Recent evidence indicates that blastocyst aneuploidy rates are higher in women with infertility or recurrent pregnancy loss compared with fertile women.[48] In addition, aneuploidy rates in oocytes have been linked to the curve of natural fertility in females,[49] whereby whole chromosome nondisjunction events are common in young girls <15 years of age and loss of cohesion resulting in inverted and deteriorated bivalent (univalents) events more prevalent in older women, the latter leading to decreased fertility due to implantation failure or miscarriage. Chromosomal meiotic pairing anomalies are more common in oocytes compared with spermatocytes[50] and are a frequent cause of female infertility/POI.[36,51]

Genetics associations with age at natural menopause, POI, and menarche

Aside from the more obvious monogenic abnormalities mentioned, there are more subtle genetic factors influencing gonadotoxic risk and fertility. Several studies have uncovered possible genetic causes influencing age at natural menopause (ANM)[52,53] as well as early menopause[54] in the general population. Certain variants of these genes, such as MCM8 and HELQ also have associations with POI, suggesting that their variants contribute to a continuum of ovarian reserve, with mild forms causing depleted ovarian reserve and severe variants causing POI.[52,55] A recent large genome-wide association study (GWAS) found 290 genetic determinants of ANM, replicated in an East Asian population, though heterogeneity of effect sizes and allele frequencies in the replication cohort were present.[56] The genetic determinates explained over 30% of the estimated heritability of ANM variance. There was a significant polygenic influence in ANM from 34 to 61 years, with the polygenic score performing significantly better than smoking status at predicting POI, which is the most robust epidemiologically associated risk factor. Those in the top 1% of the polygenic score had POI risk equivalent to those with FMR1 premutations, the leading tested monogenic cause of POI (OR 4.71 and ~5, respectively). The expression-linked genes showed the importance of DNA repair and damage response (DDR) processes and metabolic signaling networks in regulating ANM, with enrichment in genes expressed in gonadal tissue but not hypothalamic and pituitary expressed genes. The DDR pathways included 58 genes associated with apoptosis, giving evidence for ANM variation being due to variation in apoptosis following DDR. In addition, earlier age at menarche was associated with earlier ANM. This study agrees with an earlier GWAS and exome meta-analysis[57] showing ANM associated with DDR genes, POI genes, hypogonadotrophic hypogonadism genes, and the onset of puberty.

Genetic links between cancer and fertility

The possible genetic link between cancer and fertility can further complicate the study of infertility. For women, genetic variants influencing ANM have associations with hormone-sensitive cancers, likely due to variations in menopause timing, with later menopause being associated with these cancers.[56,57] Also, BRCA1[37–42] and BRCA2 mutations,[43] which are associated with breast and ovarian cancers, are also associated with POI. In addition, women with NBS, Fanconi anemia or MCM8/9 variants are at greater risk for both POI and cancer development.[14,58–60]

For men, it was found that the ERCC1 gene, which is associated with glioblastoma and oral squamous cell carcinoma cancer, was significantly associated with azoospermia.[21] In addition, the inactivation or sub-function of GPRC6A, which is associated with prostate cancer, was reported to lead to low sperm counts in adulthood.[22] Finally, a large genomic study found associations between puberty timings and the risk of breast and endometrial cancers in women and prostate cancer in men.[61]

Genetics of alkylator metabolism rates, pharmacogenomics and phenoconversion

Aside from genetic factors influencing fertility directly, there are genetic factors that influence the metabolism or bioactivation of alkylator drugs, referred to as pharmacogenomics. An individual's metabolizer state can also influence gonadal function. The most studied alkylator is cyclophosphamide (CP), which requires bioactivation by hepatic cytochrome P450 (CYP) enzymes. Individuals with single-nucleotide polymorphisms (SNPs) affecting these enzymes may therefore show a variation in their metabolism of CP. For example, around 3% of people of European ancestry and 20% of people of Japanese ancestry have loss-of-function variants in both alleles for CYP2C19 and are therefore, genotypically poor metabolizers of CP.[62] For a recent review, refer to Highley (2022).[63]

While CP is the most studied alkylator, other alkylators are used in pediatric cancer treatment and are included in the commonly utilized cyclophosphamide equivalent dose (CED) formula[64] to assess gonadotoxicity. Genetic polymorphisms are known to also affect ifosfamide and melphalan.[63] While the CED formula is convenient for calculating combined gonadotoxic risk, it is less useful from the perspective of personalized medicine, since individualized pharmacogenomic contributions to metabolism are not incorporated. A further limitation is that the CED formula does not include cisplatin or carboplatin, which are also gonadotoxic.

In addition, genes associated with the pharmacogenomics of CP fail to account for interindividual variability.[63] The discordance between genotype and phenotype, also known as phenoconversion, can further influence alkylator metabolism through co-medications, inflammation, or acquired resistance. A recent study found that half of breast cancer patients experienced CP phenoconversion, with one-third showing increased bioactivation and 15% showing decreased bioactivation between cycles, possibly due to CYP450 genetic interactions or downregulation or enzymes during periods of inflammation.[62] In addition, resistance to CP, ifosfamide, chlorambucil,[63] and cisplatin[65] can be acquired.

Genetics of DNA repair and damage response

Finally, while pharmacogenomics may in part determine the extent of gonadal damage due to cancer treatment, recovery from the gonadotoxic effects is associated with DDR genes. Unfortunately, DDR genes do not appear to be an independent factor but instead are associated with gonadal dysgenesis,[12] cancer,[42,43] ANM,[56,57] and POI,[36] as previously mentioned. Therefore, the DDR gene confounders must be properly accounted for in understanding the pharmacogenomics of fertility recovery after cancer therapy.

Aims

While the set of confounding factors is quite large, various studies have considered the pharmacogenomics of infertility induced by gonadotoxic cancer treatments.[66,67] The picture is further complicated, however, when considering children, adolescents, and young adults (CAYA). Among cancer patients, the most studied population is adults, with childhood cancers being relatively rare. Since childhood cancers differ from adult cancers in type, location, and causes, it is of interest to understand the pharmacogenomics of infertility in childhood cancer survivors, specifically.

Therefore, the aim of this research is to assess pharmacogenomic studies of long-term fertility in CAYA cancer survivors, considering the interaction of cancer treatment received and fertility-related outcomes, while also evaluating these studies in light of the various outlined confounders.

MATERIALS AND METHODS

This review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement (PRISMA) guidelines[68] and was registered with the International Prospective Register of Systematic Reviews (PROSPERO) under ID 445984. A review protocol other than what is outlined here has not been prepared.

Selection criteria

Published studies which fulfilled the following criteria were included: (i) CAYA cancer survivors, below 25 years old at the time of diagnosis, (ii) report fertility outcome measures after cancer therapy, (iii) genetics studied.

Studies were excluded if they were (i) conducted in animal models, (ii) were not published in English, (iii) editorial letters, (iv) theses.

Search strategy

Advanced searches were made in Ovid MEDLINE, Embase Classic+Embase, and PubMed. The exact search strategies utilized for each database are shown in the Supplementary Information.

Screening

The literature search results were uploaded to the systematic review software Covidence,[69] followed by the automatic identification and removal of duplicates. Three independent reviewers used a two-stage screening process to first review the titles and abstracts, and secondly assess the full text for the inclusion and exclusion criteria. Any conflicts were resolved through collaborative discussion, leading to consensus.

Data extraction

For each included article, the following data were extracted: cohort details (selection criteria, demographics, and cohort therapies) as shown in Table S1, model details (genes studied, polymorphisms studied, end points, type of models of analysis) as given in Table S2, and gene results (variant, genotype, effect, risk ratio) as given in Table S3. A second reviewer independently checked each of the extracted data points.

Quality assessment

The risk of bias was assessed by two independent reviewers using the Quality in Prognostic Studies (QUIPS) assessment scale,[70] with results given in Table 4. The QUIPS tool assesses six domains: study participation, attrition, prognostic factor measurement, outcome measurement, study confounding, statistical analysis, and reporting. Each of the six potential bias domains was rated as high, moderate, or low risk of bias by two independent reviewers, with ratings compared and a final rating determined via consensus. A summated score for overall study quality is not recommended by QUIPS, but instead, the ratings of the six domains were evaluated qualitatively, with the importance of each criterion kept in mind.

RESULTS

Study selection

The search strategy retrieved a total of 428 records. After the removal of duplicates, 326 records underwent title and abstract screening, followed by full-text screening. The secondary selection reviewed full texts to establish the relevance of the 30 articles. We used citation searching together with the literature search since the literature search only yielded review articles,[12,66,71–73] detailed in Table 1. By assessing the citations in the review articles, another review article[74] also was found.

While the literature search did not result in any articles, the citations and review articles led to our eight final studies that allowed for data extraction and analysis. The review articles themselves were excluded. The study selection process is outlined in the PRISMA chart of Figure 1.

Study characteristics

This systematic review comprises eight pharmacogenomic studies of long-term fertility in pediatric cancer survivors,[75–82] denoted S1–S8 throughout this article. Table 2 summaries the characteristics and approaches of each of these included studies.

Populations

Of the 8 included studies, 5 cohorts consisted of patients of European descent. This included two Swedish cohorts,

TABLE 1 The review articles found from the initial screening process.

First author	Year	Scope
van der Perk, M. E. M.	2022	Oncofertility in pediatric Wilms tumor patients
Mastellari, E.	2020	Genetics of female fertility
Gramatges, M. M.	2018	Genetic risks of treatment toxicities in pediatric cancer survivors
Clemens, E.	2018	Genetic variations of treatment toxicities in pediatric cancer survivors
Pinto, N.	2009	Pharmacogenetics related to cyclophosphamide toxicities

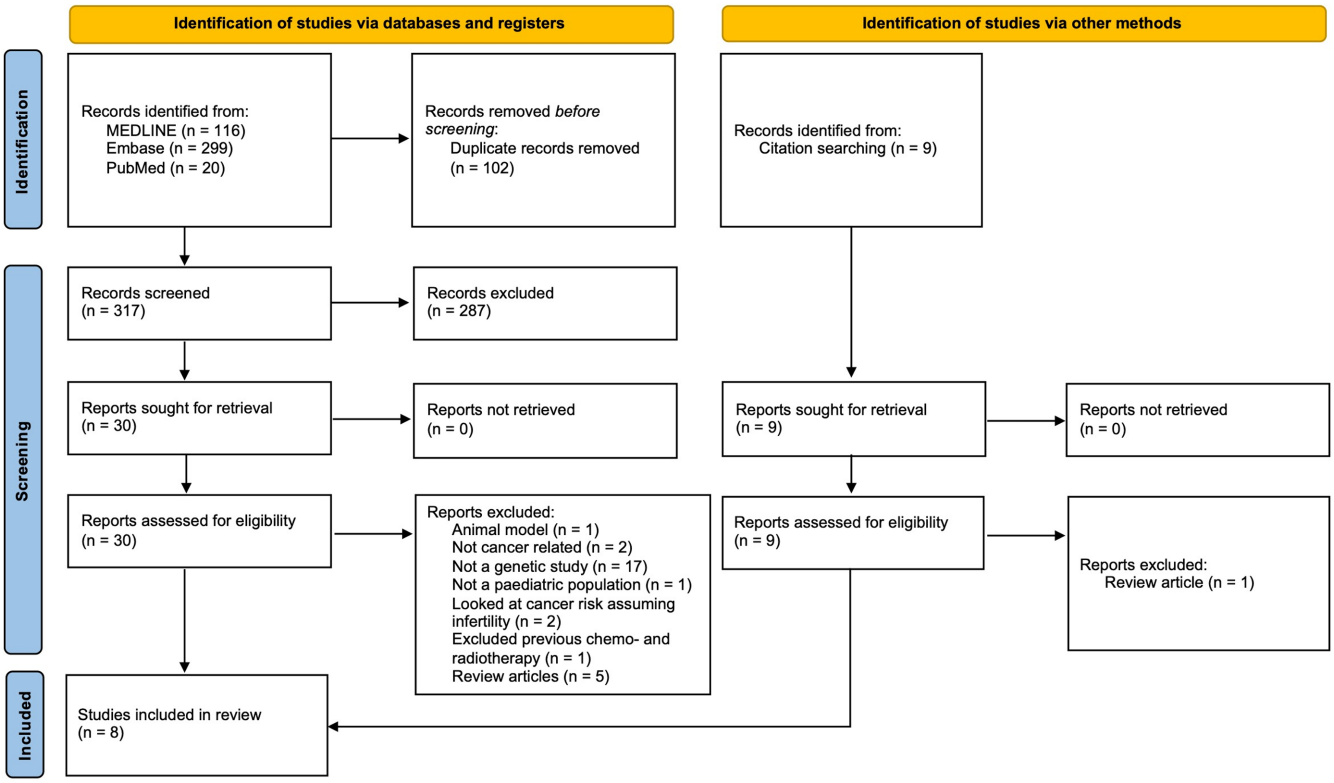

FIGURE 1 PRISMA Chart showing the systematic review process and study selection.

TABLE 2 Summary of the pharmacogenomic studies of fertility in pediatric cancer patients included in this systematic review.

Study	First author	Year	Sex	Method	Genes	Trinucleotide repeats/SNPs
S1	Eberhard, J	2004	Males	Candidate gene	AR	CAG and GGN repeats
S2	Romerius, P	2011	Males	Candidate gene	AR, ESR1, ESR2 (previously known as ERα, Erβ)	CAG and GGN repeats, 51 SNPs
S3	van Dorp, W	2013	Females	Candidate gene	ARHGEF7, MCM8, IGF2R, PCSK1, TNF, BRSK1	7 SNPs
S4	Brooke, R J	2017	Females	GWAS	NPY2R	13 SNPs
S5	Brooke, R J	2018	Females	GWAS	NPY2R, ZNF485, DAB2, UGT2A3, PRKCE, LGI4	20 SNPs
S6	van der Kooi, A-L L F	2021	Females	Candidate gene	BRSK1, FANCI, MCM8, UIMC1, RAD51	5 SNPs
S7	van der Perk, M E M	2021	Females	Candidate gene	CYP2C19, CYP3A4, CYP2B6	9 SNPs
S8	Rotz, S J	2022	Males and Females	GWAS and Whole genome replication	ST6GALNAC3, PROS1, PROS2, TNFAIP8, TNFAIP9, TNFAIP10, TNFAIP11, PELO	9 SNPs

the SWENOTECA cohort, and the Swedish Tumor Registry cohort (S1, S2); two Dutch, a cohort recruited from Erasmus MC-Sophia Children's Hospital (S3) and The Dutch Cancer Oncology Group (DCOG) (S6); and the PanCareLIFE Cohort (S6, S7 and see details in ref. [83]). The Childhood Cancer Survivor Study (CCSS) (S4, S5, S8), based in the USA, also had a large proportion of patients with European ancestry. The St. Jude Lifetime Cohort Study (SJLIFE) is an American prospective cohort analyzed in multiple studies (S4–S8)[84,85] with ethnicity only specified in one study (S5), where around 14% were Black, and 85% White.

Cohort sizes ranged from 112 to 1709 participants. Larger studies used well-established cancer survival cohorts such as SJLIFE and CCSS for recruitment and analysis. Of the 8 studies, 7 were of single-sex cohorts, 2 male (S1, S2) and 5 female (S3–S7), while only one study used a mixed-sex cohort (S8).

Cohorts did not exclude any cancer types, although S1 only included males with a diagnosis of testicular germ cell cancer. A wide variety of tumor types and diagnoses were present in the cohorts, and therapy included both having or not having alkylating agents, radiotherapy (RT), or surgery. Studies S6, S7 excluded patients with extensive gonadal toxicity.

Most of the studies included a retrospective cohort (S2–S8). S1 used a prospective cohort as well as the studies S4–S8, which included the prospective SJLIFE cohort. Studies S4–S8 were the only studies that included one or more replication cohorts.

See Table S1 for details of the cohorts in each study.

Prediction models

The pharmacogenomics studies used many heterogenous models to look at diverse different outcome measures, detailed in Table S2. The male studies considered sperm concentration and motility (S1), azoospermia (S2), and self-reported "ever siring a pregnancy" (S8) as outcome measures, looking at the effect of trinucleotide repeats in the androgen receptor (AR) gene (S1, S2), and individual SNPs of the AR, ESR1, ESR2 (S2) and PELO (S8) genes concerning chemotherapy (S1, S2, S8) and/or radiotherapy (S1, S2). S1 used chemotherapy categories according to individual protocols, S2 used dosage thresholds, and S8 considered no CED vs. any CED. Study S8 also adjusted for birth year (< vs. ≥1965) and ancestry principal components.

The female studies looked at AMH levels premature menopause (PM) defined in these studies as menopause <40 y, predicted age at menopause, or ever having a pregnancy as outcome measures. The AMH levels were considered either as categories (<vs. ≥1 μg/L (S3) or low vs. high tertiles within age groups among cohort (S6, S7), or < vs. ≥ −1.5 or −2 standard deviations within age groups among cohort (S7)) or as log-transformed continuous AMH levels (S3, S6, S7). The PM was either clinically assessed (S4, S5) or self-reported by means of a survey (S4–S7), the predicted age at menopause used population models based on AMH levels, and ever having a pregnancy was self-reported via a questionnaire. The models studied adjusted for either alkylating agent dose (AAD) using tertiles among cohort (S3), CED < vs. ≥8 g/m^2 (S5), CED in the categories 0, >0–4 g/m^2, ≥4–8 g/m^2, ≥8 g/m^2 (S6, S7), or having any CED (0 vs. >0 g/m^2) (S8), as well as radiotherapy (S3–S5). In addition, the models also adjusted for age (S3, S5–S7), birth year (< vs. ≥1965) (S8), hormone use (S6), and ancestry principal components (S5–S8).

These significant differences between studies limit direct comparisons of results.

Gene results

Of the 29 genes studied, there were 9 genes which had significant results, even if they could not be replicated either in the original study, a replication cohort, or a different study. See Table S3 for details of the gene results and Table 3 for a summary of attempts at replication.

Only genes NPY2R and BRSK1 had significant findings in two different studies. However, in the case of NPY2R, S4 is a published conference abstract with few details, with S5 being from the same group. Regarding the BRSK1 gene, different SNPs were shown to be significant between S3 and S6, and furthermore in S6, there were conflicting effect directions for the SNPs. Nevertheless, if we look within studies using replication cohorts, then CYP3A4 was replicated in S7's replication cohort and meta-analysis; the NPY2R results in S5 were replicated in its replication cohort as a haplotype of 4 SNPs, although one of the SNPs was different between the discovery and replication cohorts; and CYP2B6 failed to be replicated in S7's replication cohort but yielded a significant result in the meta-analysis combining both the discovery and replication cohort. Pharmacogenomic results are therefore still quite unclear.

Table S2 includes all significant results for each SNP studied, or if the SNP was not found to be significant, the model that gives the smallest *p*-value for a given SNP.

Risk of bias – QUIPS

The risk of bias assessment results are summarized in Table 4, revealing a significant risk of bias assessed for a majority of the studies. We note that S4 was a published conference abstract, and as such was assessed as high risk of bias for all categories due to having insufficient details to make a proper assessment. We therefore exclude S4 from the discussion below.

Studies S2, S3, S5 had a low risk of bias for the study participation domain. The recruitment of these cohorts was well described, the study populations were appropriate, and participation was adequate for analysis. In S1, participation at different timepoints differed between participants, even if the participants were well described and appropriate, while in S6 the replication cohort lacked

TABLE 3 Attempts at replication of significant findings in studies S1–S8.

Gene studied	Studies showing significant result	Replication attempts
AR	S1	Failed to replicate in S2
ESR1	S2	Not replicated
BRSK1	S3, S6	Result from S3 – not replicated Result from S6 – replicated in replication cohort of S6, but results and replication had conflicting protective vs. predisposing effects
NPY2R	S4, S5	Result from S4 – replicated in replication cohort of S4 but without details of replication or of SNPs used in haplotype(s) Result from S5 – replicated in replication cohort of S5 but with modified haplotype
FANCI	S6	Failed replication in both replication cohorts of S6
CYP2C19	S7	Failed in replication cohort and in meta-analysis of S7
CYP3A4	S7	Replicated in replication cohort and in meta-analysis of S7
CYP2B6	S7	Failed in replication cohort but replicated in meta-analysis S7
TNFAIP8	S8	Failed in replication cohort of S8

TABLE 4 Risk of bias assessment for each of the 6 QUIPS domains, classified as low, medium, or high, plus an overall assessment for each of the studies S1–S8.

Study	Study Participation	Study Attrition	Prognostic Factor Measurement	Outcome Measurement	Study Confounding	Statistical Analysis and Reporting	Overall Score
S1	Medium	High	High	Low	Medium	High	High
S2	Low	Low	Low	Low	Medium	Medium	Low
S3	Low	Low	Medium	Low	High	Low	Medium
S4	High*	High*	High*	High*	High*	High*	High*
S5	Low	Medium	Medium	Medium	High	Medium	High
S6	Medium	High	Low	High	High	Medium	High
S7	High	High	Low	High	High	Medium	High
S8	High	High	High	Medium	High	Medium	High

*Conference abstract lacking in details required for assessment.

recruitment details and survivorship, inclusion criteria were different between cohorts, yielding a risk of bias of medium for S1 and S6. The studies S7, S8 lacked recruitment details, yielding a high risk of bias.

Articles which failed to describe the loss to follow-up or dropout rates were attributed a high risk of bias for the study attrition domain (S6–S8). Study S5 excluded those who had acute ovarian failure (AOF), who were also more likely to have received a higher dose of radiotherapy and less likely to have had lymphoma, yielding a medium risk of bias. Study S1 received a high risk of bias since 30% of participants did not give follow-up samples, and those who dropped out were not compared with those who remained.

Prognostic factor measurements were well described across the reviewed studies; however, we rated as a high risk of bias S1 and S8. In S1, only the first 83 recruited participants had DNA analysis, those who had orchidectomy,

and data were not given to compare how those having DNA analysis differed from those who did not. In S8, the discovery cohort used GWAS while the replication cohort used whole genome sequencing, CED was not stratified according to dosage levels, and an arbitrary birth year cutoff of 1965 was used to group the cohorts into two age brackets. Studies S2, S6, S7 were given low risk of bias assessments in this domain, even though S6, S7 did not further stratify CED risk according to pubertal status (see risk tables in ref. [5]). Study S3 was given a medium risk of bias since AAD, defined by tertiles within the cohort, was used instead of CED and because they included participants up to the age of 50 who may have already naturally reached menopause without the effect of cancer treatment. Study S5 was also given a medium risk of bias since the replication cohort used a different SNP in the haplotype considered.

Generally, the study outcomes were defined clearly. However, a high risk of bias was attributed to S6, S7, both of which used AMH tertiles defined within their cohorts rather than compared against population models for defining controls and cases (S7 also considered AMH category levels based on standard deviation thresholds from within the cohorts), they did not define controls for those ≥40 years, and they relied on participant surveys for determining PM. Study S8 also relied on participant questionnaires to determine having or siring a pregnancy, for which a medium risk of bias was attributed.

Six studies (S3–S8) were assessed with a high risk of bias for the study confounding domain. Study S3 included AMH levels from pregnant women and those taking contraceptive pills (55.1%) as well as those on hormone replacement therapy (4%), which formed a majority of the study cohort and was uncorrected for. Study S5 excluded those with AOF, but nevertheless defined PM as having amenorrhea >6 months (in association with estradiol lower than 17 pg/mL and FSH higher than 30 IU/L) with the relationship between AOF and PM not clarified. Also, S5 only looked at PM without consideration of age as a confounder, leading to them finding those with PM being older than those without PM. In addition, S5 did not consider RT dosages (only yes/no categories) and the excluded participants were more likely to not have survived and/or had more RT. Study S6 excluded those at the highest risk of infertility (those who had allogenic stem cell transplantation, total body irradiation, bilateral ovary-exposing RT, cranial and/or craniospinal RT), and hormone use was not available in the replication cohort. In S7, those with higher gonadotoxic treatment were excluded, and hormone therapy or contraceptive use was not available in the discovery cohort, so it could not be adjusted for. In addition, the replication cohort was older, with a median above the third quartile of the discovery cohort. In S8, hormone treatment and contraceptive use were not described, and the questionnaire did not ask about attempts at assisted reproductive technology (ART), did not consider multiple pregnancies, excluded those who never married, and did not ask whether the marriage was heterosexual. Ethnicities were not specified in studies S6–S8. Study S1 received a medium risk of bias since it did not account for age and its only significant pharmacogenomics result was for a small sub-cohort which was not compared against the rest of the cohort. Study S2 also received a medium risk of bias since it did not adjust for age.

Regarding the statistical analysis domain, Study S1 received a high risk of bias since it was not clear why the particular specific category of participant that yielded a significant result was chosen (those with 3–4 cycles of chemotherapy 12–24 months after treatment) and the *p*-value was not adjusted accordingly. Study S2 received a medium risk of bias since it only presented significant results and did not give details about the SNPs that it did not find significant, and *p*-value adjustment was not used in selecting SNPs. Study S5 received a medium risk of bias since the results in Table 2 appear to be incorrect, in some cases, the number of participants being less than the cohort total, and in others being above the total number in their study cohort. Study S6 was given a high risk of bias since the BRSK1 SNP had both protective (Models 1, 3, 4 in Table S2) and predisposing (Models 4–8 in Table S2) pharmacogenomic effect, even swapping direction (Model 4) depending on the interaction with CED term, while the results were presented only as a reduced ovarian function effect (the FANCI SNP also swapped direction but was left out of the abstract). In S7, the AMH assay limit of quantitation was 0.03 μg/L and those having the lowest AMH levels were given this value, showing a large spike in the frequency plot versus the standard deviation (with respect to the natural log of AMH) in their Figure S3 and potentially biasing the standard deviation (SD) calculations used in Models 5–6 described in Table S2. In addition to using AMH tertiles within their cohort to define controls and cases, S7 also used SD thresholds but comparing those less than the threshold (either −1.5 or −2 SD) to all those above, and it is unclear why these thresholds were chosen and why all those above were considered controls, leading to a medium risk of bias. For S8, which received a medium risk of bias assessment, they wished to study SNPs they call suggestive with *p*-values $<10^{-7}$; however, only one of the 8 SNPs they studied fulfilled this criterion. In addition, missing data in S8 are only mentioned via a table caption and not noted in the main text, and it is unclear how the analysis accounted for the data gaps.

By considering each of the 6 domains, we have given the overall risk of bias assessments as low for S2, medium

for S3, and high for the other studies, as can be seen in Table 4.

DISCUSSION

One of the first things to notice in our results is the paucity of studies on males. The two oldest studies from 2004 (S1) and 2011 (S2) are the only studies specifically dedicated to male childhood cancer survivors, with the only other study (S8) being a combined male and female study.

Secondly, comparing the genes found in the studies, it is clear that there is very little overlap in results between studies. In fact, S8 failed to replicate its findings in its replication cohort, and S2 failed to replicate S1. The only gene that was replicated in separately published studies is BRSK1 (S3, S6), although for different SNPs (S4 and S5 replicated findings, but the former is a published conference abstract later expanded in the latter). However, several studies found tentative replication within their replication cohort, NPY2R and associated haplotype (S4, S5) but with modified haplotype in replication cohort (S5), BRSK1 (S6) although with an opposite effect in a replication cohort, and CYP2B6 (S7) which failed replication in its replication cohort but was replicated in the meta-analysis. It should be noted that BRSK1 is associated with POI and ANM independent of cancer status.[86]

The interplay between genetics and cancer predisposition also presents a complex correlation. For example, cancer patients who carry mutations in DNA repair genes, for example BRCA1, BRCA2, may be more susceptible to alkylator damage,[37–43] and patients with variants of DNA repair genes, such as NBN, MCM8/9 likely have elevated risk of infertility and cancer independent of therapy.[14,58–60] Repair mechanisms are limited, increasing cytotoxicity and the risk of further mutagenesis. The unique nature of each clinical case highlights the need for a well-represented cohort for analysis, including heterogeneity with respect to ethnicity and disease state.

Third, it is important to point out the heterogeneity of both the cohorts and the phenotyping. The complex nature of gametogenesis introduces a wide scope of possible causal relationships. At present, there is no standard method of analysis to measure defined outcomes. A number of studies have assessed different polymorphisms, for example, hormone levels, PM, and having or siring a pregnancy. This multifaceted approach creates complexity in the literature and existing study results lack robustness for clinical implementation. Certainly, further studies involving next-generation sequencing and prospective, large cohorts, with outcome measures that can be easily replicated are needed.

Being able to identify patients who are at the highest risk of infertility has important implications for how patients are counseled prior to treatment, particularly for pre-pubertal patients. While post-pubertal patients have access to more established fertility preservation options, such as sperm and egg freezing, pre-pubertal patients can currently only be offered testicular or ovarian tissue cryopreservation. Both of which are recent techniques, with testicular tissue cryopreservation still considered experimental. Due to the novel nature of these procedures, only certain centers offer them to pre-pubertal patients, and only to those at higher risk of infertility. Therefore, if those at higher risk of infertility can be more accurately identified, patients and their families can make better informed choices regarding tissue cryopreservation, which may help to minimize potential future decisional regret.

Finally, since fertility is a long-term outcome in childhood cancer survivors, one of the best methods of studying prevalence and risk-factors by means of longitudinal cohort studies. There have been a number of childhood cancer survivor cohort studies around the world: the CCSS, the St. Jude Lifetime Cohort Study (SJLIFE), the Dutch Childhood Oncology Group Long-term Effects After Childhood Cancer (DCOG LATER), PanCareLIFE, the Swedish and Norwegian Testicular Cancer Group (SWENOTECA), Adult Life after Childhood Cancer in Scandinavia (ALiCCS), Childhood, Adolescent, Young Adult Cancer Survivors (CAYACS), the British Childhood Cancer Survivor Study (BCCSS), the Swiss Childhood Cancer Survivor Study (SCCSS), the French Childhood Cancer Survivor Study (FCCSS), the French Childhood Cancer Survivor Study for Leukemia (LEA), and the Italian Off-Therapy Registry (OTR).[87,88] Nevertheless, it appears that only half of these cohort studies have studied fertility as a survivorship issue. We also note that two of the studies considered in this review appear not to come from longitudinal studies: S2 studies retrospective patients from the Region of Southern Sweden in the Swedish Tumor Registry, and S3 is a retrospective study from the Erasmus Medical Center-Sophia Children's Hospital. Of the cohort studies that have addressed fertility, all except for OTR's[89] have looked at the pharmacogenomics of cancer treatment-induced infertility, showing the importance placed on considering genetics.

With the exception of SWENOTECA and SJLIFE, literature regarding the risk prediction of infertility in childhood cancer survivors has taken a retrospective approach. Early research has perhaps been driven by discovery. The cohorts analyzed may be biased by institutional practices, when recruitment has been conducted

by convenience sampling (S3) or using a national registry (S1, S2, S6), impeding replication and failing to account for the inextricable link between pharmacogenomics and ethnicity.[90] Now with increasing evidence of known genes associated with infertility risk, the need for prospective research is highlighted, which allows for the validation of a predetermined and specific panel of genetic variants.

Limitations

One potential limitation of this study is that we excluded pharmacogenomics studies of cyclophosphamide-induced infertility in lupus patients. Two studies[91,92] include CAYA patients (plus adults) receiving cyclophosphamide (CP) for treating lupus and find CYP2C19*2 to be significantly protective for POI. One study[93] was not powered sufficiently to find a significant effect, but the results suggested that having CYP2C19*1 or CYP2C19*1/*2 instead of CYP2C19*2 increases ovarian toxicity. Another study[94] looked at the pharmacokinetics of CP in lupus patients aged 12–60, finding that variants CYP2C19*2 and CYP2B6-750T > C accounted for most of the variation in 4-OH-CPA plasma concentration (21.5% and 23.6%, respectively).

Of the genes studied in these lupus cohorts, only CYP2C19 and CYP2B6 have also been studied in the CAYA cancer population of this systematic review, with the only common SNP being CYP2C19*2 considered in S7 and in the four lupus studies.[91–94] Two lupus studies found that CYP2C19*2 significantly lowers the odds of POI,[91,92] and while S7 also found a protective effect in the logistic regression odds ratio based on the lowest versus highest AMH tertile per age group, the result did not reach significance (*p*-value 0.191, details in Table S3). This suggests that perhaps some of the underlying CP pharmacogenomic ovarian toxicities affect lupus and cancer patients similarly.

Another similar potential limitation is the exclusion of adult pharmacogenomics studies of cancer therapy-induced infertility. The results from Ruth (2021) and Day (2015) have uncovered links between age at menarche and ANM,[56,57] suggesting longitudinal effects. The study by Gruhn (2019) has shown a mixed picture, where whole chromosome nondisjunction events grow in number linearly with age starting as young as 15 years, while centrometric, fully inverted and deteriorated bivalent events are more prevalent in older women.[49] While it is unclear how to untangle the potentially different effects in adult and CAYA populations, we note that studies S1–S3, S6, S7 have used a candidate gene approach that used fertility-associated genes from adult populations as the starting point of their studies. It is unclear to what extent the results of these studies were hindered or helped by using these genes as their starting points.

It is also important to note that the search strategy used did not directly yield any of the relevant studies. It was only in reading cited references (and cited references within them) that they were eventually found. We recommend that a future systematic review carefully assess the search strategy used here and make adjustments to ensure the retrieval of key articles (S1–S8).

Future directions

Only one of the studies performed whole genomic sequencing (S8), and only within the replication cohort, the other studies used GWAS or candidate gene approaches. Since GWAS and candidate gene studies inevitably miss relevant genetic polymorphisms,[95] it will be important for future studies to utilize whole genome sequencing.

In addition, little is known of the phenoconversion and co-medication effects of many chemotherapy drugs administered to pediatric cancer patients. This is an important confounder of any pharmacogenomics analysis, and it will be important to adjust for it in any future study, requiring separate dedicated analyses.

CONCLUSIONS

This systematic review evaluated genetic polymorphisms associated with infertility in CAYA cancer survivors. Eight articles included in this review (S1–S8) described genetic variants with either predisposing or protective effects on infertility. The findings, however, remain conflicting as many studies named a disparate set of genes (Tables 2 and 3). In addition, the effect of some of the significant SNPs changed direction, depending on CED levels or cohort. The heterogeneity of these results makes it extremely difficult to draw clinical conclusions. The impact of these results may be further limited due to small cohort sizes (S1–S3, S6), a lack of properly matched control groups (S6, S7), and variations in outcome measures and models of analysis, all potential key factors which may contribute to the difficulty and inability to replicate findings.

Pharmacogenomic studies of treatment-induced infertility in childhood cancer survivors are still in their infancy. The implementation of findings is further exacerbated by the lack of standardized approaches, the lack of agreement of results, and the lack of replication. In addition, male focused studies seem to be lacking in comparison to female studies. Given the large number of potential confounding effects in this area of study, larger, prospective

longitudinal studies with an unbiased genome-wide focus will be important to uncover significant results that will be consistently replicated.

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REFERENCES

1. Johnston WT, Erdmann F, Newton R, Steliarova-Foucher E, Schüz J, Roman E. Childhood cancer: estimating regional and global incidence. *Cancer Epidemiol*. 2021;71(Pt B):101662.
2. Bhakta N, Force LM, Allemani C, et al. Childhood cancer burden: a review of global estimates. *Lancet Oncol*. 2019;20(1):e42-e53.
3. Jayasuriya S, Peate M, Allingham C, et al. Satisfaction, disappointment and regret surrounding fertility preservation decisions in the paediatric and adolescent cancer population. *J Assist Reprod Genet*. 2019;36(9):1805-1822.
4. Li N, Jayasinghe Y, Kemertzis MA, Moore P, Peate M. Fertility preservation in pediatric and adolescent oncology patients: the decision-making process of parents. *J Adolesc Young Adult Oncol*. 2017;6(2):213-222.
5. Meacham LR, Burns K, Orwig KE, Levine J. Standardizing risk assessment for treatment-related gonadal insufficiency and infertility in childhood adolescent and young adult cancer: the Pediatric Initiative Network risk stratification system. *J Adolesc Young Adult Oncol*. 2020;9(6):662-666.
6. Kelsey TW, Anderson RA, Wright P, Nelson SM, Wallace WHB. Data-driven assessment of the human ovarian reserve. *Mol Hum Reprod*. 2012;18(2):79-87.
7. Jayasinghe YL, Wallace WHB, Anderson RA. Ovarian function, fertility and reproductive lifespan in cancer patients. *Expert Rev Endocrinol Metab*. 2018;13(3):125-136.
8. Aznavour Y, Navarrete F, Badreddine J, et al. Semen parameter variability among users of at-home sperm testing kits. *BMC Urol*. 2022;22(1):184.
9. Gravholt CH, Viuff M, Just J, et al. The changing face of turner syndrome. *Endocr Rev*. 2023;44(1):33-69.
10. Parizot E, Dard R, Janel N, Vialard F. Down syndrome and infertility: what support should we provide? *J Assist Reprod Genet*. 2019;36(6):1063-1067.
11. Katari S, Aarabi M, Kintigh A, et al. Chromosomal instability in women with primary ovarian insufficiency. *Hum Reprod*. 2018;33(3):531-538.
12. Mastellari E, La Marca A. Genetic conditions impairing female fertility. *Panminerva Med*. 2020;62(4):260-267.
13. Yatsenko SA, Rajkovic A. Genetics of human female infertility. *Biol Reprod*. 2019;101(3):549-566.
14. Tucker EJ, Grover SR, Robevska G, van den Bergen J, Hanna C, Sinclair AH. Identification of variants in pleiotropic genes causing "isolated" premature ovarian insufficiency: implications for medical practice. *Eur J Hum Genet*. 2018;26(9):1319-1328.
15. Fouquet B, Pawlikowska P, Caburet S, et al. A homozygous FANCM mutation underlies a familial case of non-syndromic primary ovarian insufficiency. *eLife*. 2017;6:e30490.
16. Cunniff C, Jones KL, Benirschke K. Ovarian dysgenesis in individuals with chromosomal abnormalities. *Hum Genet*. 1991;86(6):552-556.
17. Russell P, Altshuler G. The ovarian dysgenesis of trisomy 18. *Pathology*. 1975;7(2):149-155.
18. Hojager B, Peters H, Byskov AG, Faber M. Follicular development in ovaries of children with Down's syndrome. *Acta Paediatr Scand*. 1978;67(5):637-643.
19. Grinspon RP, Bergada I, Rey RA. Male hypogonadism and disorders of sex development. *Front Endocrinol (Lausanne)*. 2020;11:211.
20. Grinspon RP, Rey RA. When hormone defects cannot explain it: malformative disorders of sex development. *Birth Defects Res C Embryo Today*. 2014;102(4):359-373.
21. Ji G, Gu A, Xia Y, et al. ERCC1 and ERCC2 polymorphisms and risk of idiopathic azoospermia in a Chinese population. *Reprod Biomed Online*. 2008;17(1):36-41.

22. De Toni L, Di Nisio A, Speltra E, et al. Polymorphism rs2274911 of GPRC6A as a novel risk factor for testis failure. *J Clin Endocrinol Metab*. 2016;101(3):953-961.
23. Krausz C, Riera-Escamilla A, Moreno-Mendoza D, et al. Genetic dissection of spermatogenic arrest through exome analysis: clinical implications for the management of azoospermic men. *Genet Med*. 2020;22(12):1956-1966.
24. Greither T, Schumacher J, Dejung M, et al. Fertility relevance probability analysis shortlists genetic markers for male fertility impairment. *Cytogenet Genome Res*. 2020;160(9):506-522.
25. Kuroda S, Usui K, Sanjo H, et al. Genetic disorders and male infertility. *Reprod Med Biol*. 2020;19(4):314-322.
26. Marzouni ET, Harchegani AB, Layali I. Chromosomal aneuploidies and associated rare genetic syndromes involved in male infertility. *J Men's Health*. 2021;17(3):7-17.
27. Aston KI. Genetic susceptibility to male infertility: news from genome-wide association studies. *Andrology*. 2014;2(3):315-321.
28. Skakkebaek NE, Rajpert-de Meyts E, Buck Louis GM, et al. Male reproductive disorders and fertility trends: influences of environment and genetic susceptibility. *Physiol Rev*. 2016;96(1):55-97.
29. Juberg RC, Hayes ML. Critical region hypothesis: primary amenorrhea in an 18-year-old woman with a complex translocation (X;2;8). *Am J Med Genet Suppl*. 1987;3:145-150.
30. Krauss CM, Turksoy RN, Atkins L, McLaughlin C, Brown LG, Page DC. Familial premature ovarian failure due to an interstitial deletion of the long arm of the X chromosome. *N Engl J Med*. 1987;317(3):125-131.
31. Baronchelli S, Villa N, Redaelli S, et al. Investigating the role of X chromosome breakpoints in premature ovarian failure. *Mol Cytogenet*. 2012;5(1):32.
32. Lacombe A, Lee H, Zahed L, et al. Disruption of POF1B binding to nonmuscle Actin filaments is associated with premature ovarian failure. *Am J Hum Genet*. 2006;79(1):113-119.
33. Desai S, Rajkovic A. Genetics of reproductive aging from gonadal dysgenesis through menopause. *Semin Reprod Med*. 2017;35(2):147-159.
34. Szeliga A, Zysnarska A, Szklarska Z, et al. A case of premature ovarian insufficiency in Nijmegen breakage syndrome patient and review of literature. From gene mutation to clinical management. *Gynecol Endocrinol*. 2019;35(11):999-1002.
35. Wang F, Guo S, Li P. Two novel mutations in the MCM8 gene shared by two Chinese siblings with primary ovarian insufficiency and short stature. *Mol Genet Genomic Med*. 2020;8(9):e1396.
36. Tucker EJ, Bell KM, Robevska G, et al. Meiotic genes in premature ovarian insufficiency: variants in HROB and REC8 as likely genetic causes. *Eur J Hum Genet*. 2022;30(2):219-228.
37. Rzepka-Gorska I, Tarnowski B, Chudecka-Głaz A, Górski B, Zielińska D, Tołoczko-Grabarek A. Premature menopause in patients with BRCA1 gene mutation. *Breast Cancer Res Treat*. 2006;100(1):59-63.
38. Titus S, Li F, Stobezki R, et al. Impairment of BRCA1-related DNA double-strand break repair leads to ovarian aging in mice and humans. *Sci Transl Med*. 2013;5(172):172ra21.
39. Phillips KA, Collins IM, Milne RL, et al. Anti-Mullerian hormone serum concentrations of women with germline BRCA1 or BRCA2 mutations. *Hum Reprod*. 2016;31(5):1126-1132.
40. de la Noval BD. Potential implications on female fertility and reproductive lifespan in BRCA germline mutation women. *Arch Gynecol Obstet*. 2016;294(5):1099-1103.
41. Giordano S, Garrett-Mayer E, Mittal N, et al. Association of BRCA1 mutations with impaired ovarian reserve: connection between infertility and breast/ovarian cancer risk. *J Adolesc Young Adult Oncol*. 2016;5(4):337-343.
42. Oktay K, Kim JY, Barad D, Babayev SN. Association of BRCA1 mutations with occult primary ovarian insufficiency: a possible explanation for the link between infertility and breast/ovarian cancer risks. *J Clin Oncol*. 2010;28(2):240-244.
43. Caburet S, Heddar A, Dardillac E, et al. Homozygous hypomorphic BRCA2 variant in primary ovarian insufficiency without cancer or Fanconi anaemia trait. *J Med Genet*. 2021;58:125-134.
44. Tassone F, Iong K, Tong TH, et al. FMR1 CGG allele size and prevalence ascertained through newborn screening in the United States. *Genome Med*. 2012;4(12):100.
45. Qin Y, Jiao X, Simpson JL, Chen ZJ. Genetics of primary ovarian insufficiency: new developments and opportunities. *Hum Reprod Update*. 2015;21(6):787-808.
46. Venturella R, De Vivo V, Carlea A, et al. The genetics of non-syndromic primary ovarian insufficiency: a systematic review. *Int J Fertil Steril*. 2019;13(3):161-168.
47. Tucker EJ, Grover SR, Bachelot A, Touraine P, Sinclair AH. Premature ovarian insufficiency: new perspectives on genetic cause and phenotypic spectrum. *Endocr Rev*. 2016;37(6):609-635.
48. Kort JD, McCoy RC, Demko Z, Lathi RB. Are blastocyst aneuploidy rates different between fertile and infertile populations? *J Assist Reprod Genet*. 2018;35(3):403-408.
49. Gruhn JR, Zielinska AP, Shukla V, et al. Chromosome errors in human eggs shape natural fertility over reproductive life span. *Science*. 2019;365(6460):1466-1469.
50. Speed RM. The possible role of meiotic pairing anomalies in the atresia of human fetal oocytes. *Hum Genet*. 1988;78(3):260-266.
51. Huang C, Guo T, Qin Y. Meiotic recombination defects and premature ovarian insufficiency. *Front Cell Dev Biol*. 2021;9:652407.
52. Stolk L, Zhai G, van Meurs JBJ, et al. Loci at chromosomes 13, 19 and 20 influence age at natural menopause. *Nat Genet*. 2009;41(6):645-647.
53. He C, Kraft P, Chasman DI, et al. A large-scale candidate gene association study of age at menarche and age at natural menopause. *Hum Genet*. 2010;128(5):515-527.
54. Perry JR, Corre T, Esko T, et al. A genome-wide association study of early menopause and the combined impact of identified variants. *Hum Mol Genet*. 2013;22(7):1465-1472.
55. Stolk L, Perry JRB, Chasman DI, et al. Meta-analyses identify 13 loci associated with age at menopause and highlight DNA repair and immune pathways. *Nat Genet*. 2012;44(3):260-268.
56. Ruth KS, Day FR, Hussain J, et al. Genetic insights into biological mechanisms governing human ovarian ageing. *Nature*. 2021;596(7872):393-397.
57. Day FR, Ruth KS, Thompson DJ, et al. Large-scale genomic analyses link reproductive aging to hypothalamic signaling, breast cancer susceptibility and BRCA1-mediated DNA repair. *Nat Genet*. 2015;47(11):1294-1303.
58. Chrzanowska KH, Szarras-Czapnik M, Gajdulewicz M, et al. High prevalence of primary ovarian insufficiency in girls and young women with Nijmegen breakage syndrome: evidence from a longitudinal study. *J Clin Endocrinol Metab*. 2010;95(7):3133-3140.
59. Helderman NC, Terlouw D, Bonjoch L, et al. Molecular functions of MCM8 and MCM9 and their associated pathologies. *iScience*. 2023;26(6):106737.

60. Zhao J, Zhang Y, Li W, et al. Research progress of the Fanconi anemia pathway and premature ovarian insufficiency dagger. *Biol Reprod*. 2023;109(5):570-585.
61. Day FR, Thompson DJ, Helgason H, et al. Genomic analyses identify hundreds of variants associated with age at menarche and support a role for puberty timing in cancer risk. *Nat Genet*. 2017;49(6):834-841.
62. Helsby N, Yong M, Burns K, Findlay M, Porter D. Cyclophosphamide bioactivation pharmacogenetics in breast cancer patients. *Cancer Chemother Pharmacol*. 2021;88(3):533-542.
63. Highley MS, Landuyt B, Prenen H, Harper PG, de Bruijn EA. The nitrogen mustards. *Pharmacol Rev*. 2022;74(3):552-599.
64. Green DM, Nolan VG, Goodman PJ, et al. The cyclophosphamide equivalent dose as an approach for quantifying alkylating agent exposure: a report from the Childhood Cancer Survivor Study. *Pediatr Blood Cancer*. 2014;61(1):53-67.
65. Ghosh S. Cisplatin: the first metal based anticancer drug. *Bioorg Chem*. 2019;88:102925.
66. Pinto N, Ludeman SM, Dolan ME. Drug focus: pharmacogenetic studies related to cyclophosphamide-based therapy. *Pharmacogenomics*. 2009;10(12):1897-1903.
67. Oktay KH, Bedoschi G, Goldfarb SB, et al. Increased chemotherapy-induced ovarian reserve loss in women with germline BRCA mutations due to oocyte deoxyribonucleic acid double strand break repair deficiency. *Fertil Steril*. 2020;113(6):1251-1260.e1.
68. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.
69. Veritas Health Innovation. *Covidence Systematic Review Software*. Accessed February 05, 2024. 2023; www.covidence.org.
70. Hayden JA, van der Windt DA, Cartwright JL, Côté P, Bombardier C. Assessing bias in studies of prognostic factors. *Ann Intern Med*. 2013;158(4):280-286.
71. van der Perk MEM, Cost NG, Bos AME, et al. White paper: Oncofertility in pediatric patients with Wilms tumor. *Int J Cancer*. 2022;151(6):843-858.
72. Gramatges MM, Bhatia S. Evidence for genetic risk contributing to long-term adverse treatment effects in childhood cancer survivors. *Annu Rev Med*. 2018;69:247-262.
73. Clemens E, van der Kooi ALF, Broer L, et al. The influence of genetic variation on late toxicities in childhood cancer survivors: a review. *Crit Rev Oncol Hematol*. 2018;126:154-167.
74. Lambertini M, Goldrat O, Toss A, et al. Fertility and pregnancy issues in BRCA-mutated breast cancer patients. *Cancer Treat Rev*. 2017;59:61-70.
75. Eberhard J, Ståhl O, Giwercman Y, et al. Impact of therapy and androgen receptor polymorphism on sperm concentration in men treated for testicular germ cell cancer: a longitudinal study. *Hum Reprod*. 2004;19(6):1418-1425.
76. Romerius P, Giwercman A, Moëll C, et al. Estrogen receptor alpha single nucleotide polymorphism modifies the risk of azoospermia in childhood cancer survivors. *Pharmacogenet Genomics*. 2011;21(5):263-269.
77. van Dorp W, van den Heuvel-Eibrink MM, Stolk L, et al. Genetic variation may modify ovarian reserve in female childhood cancer survivors. *Hum Reprod*. 2013;28(4):1069-1076.
78. Brooke RJ, Chemaitilly W, Wilson CL, et al. A high-risk genetic profile for premature menopause (PM) in childhood cancer survivors (CCS) exposed to gonadotoxic therapy: a report from the St. Jude Lifetime Cohort (SJLIFE) and Childhood Cancer Survivor Study (CCSS). *J Clin Oncol*. 2017;35(15_suppl):10502.
79. Brooke RJ, Im C, Wilson CL, et al. A high-risk haplotype for premature menopause in childhood cancer survivors exposed to gonadotoxic therapy. *J Natl Cancer Inst*. 2018;110(8):895-904.
80. van der Kooi ALF, van Dijk M, Broer L, et al. Possible modification of BRSK1 on the risk of alkylating chemotherapy-related reduced ovarian function. *Hum Reprod*. 2021;36(4):1120-1133.
81. van der Perk MEM, Broer L, Yasui Y, et al. Effect of genetic variation in CYP450 on gonadal impairment in a European cohort of female childhood cancer survivors, based on a candidate gene approach: results from the PanCareLIFE Study. *Cancers (Basel)*. 2021;13(18):4598.
82. Rotz SJ, Worley S, Hu B, et al. Genome-wide association study of pregnancy in childhood cancer survivors: a report from the Childhood Cancer Survivor Study. *Cancer Epidemiol Biomarkers Prev*. 2022;31(9):1858-1862.
83. van der Kooi ALF, Clemens E, Broer L, et al. Genetic variation in gonadal impairment in female survivors of childhood cancer: a PanCareLIFE study protocol. *BMC Cancer*. 2018;18(1):930.
84. Hudson MM, Ness KK, Gurney JG, et al. Clinical ascertainment of health outcomes among adults treated for childhood cancer. *JAMA*. 2013;309(22):2371-2381.
85. Hudson MM, Ehrhardt MJ, Bhakta N, et al. Approach for classification and severity grading of long-term and late-onset health events among childhood cancer survivors in the St. Jude Lifetime Cohort. *Cancer Epidemiol Biomarkers Prev*. 2017;26(5):666-674.
86. Qin Y, Sun M, You L, et al. ESR1, HK3 and BRSK1 gene variants are associated with both age at natural menopause and premature ovarian failure. *Orphanet J Rare Dis*. 2012;7:5.
87. Winther JF, Kenborg L, Byrne J, et al. Childhood cancer survivor cohorts in Europe. *Acta Oncol*. 2015;54(5):655-668.
88. Erdmann F, Frederiksen LE, Bonaventure A, et al. Childhood cancer: survival, treatment modalities, late effects and improvements over time. *Cancer Epidemiol*. 2021;71(Pt B):101733.
89. Pivetta E, Maule MM, Pisani P, et al. Marriage and parenthood among childhood cancer survivors: a report from the Italian AIEOP off-therapy registry. *Haematologica*. 2011;96(5):744-751.
90. Ortega VE, Meyers DA. Pharmacogenetics: implications of race and ethnicity on defining genetic profiles for personalized medicine. *J Allergy Clin Immunol*. 2014;133(1):16-26.
91. Takada K, Arefayene M, Desta Z, et al. Cytochrome P450 pharmacogenetics as a predictor of toxicity and clinical response to pulse cyclophosphamide in lupus nephritis. *Arthritis Rheum*. 2004;50(7):2202-2210.
92. Singh G, Saxena N, Aggarwal A, Misra R. Cytochrome P450 polymorphism as a predictor of ovarian toxicity to pulse cyclophosphamide in systemic lupus erythematosus. *J Rheumatol*. 2007;34(4):731-733.
93. Ngamjanyaporn P, Thakkinstian A, Verasertniyom O, et al. Pharmacogenetics of cyclophosphamide and CYP2C19 polymorphism in Thai systemic lupus erythematosus. *Rheumatol Int*. 2011;31(9):1215-1218.
94. Shu W, Guan S, Yang X, et al. Genetic markers in CYP2C19 and CYP2B6 for prediction of cyclophosphamide's 4-hydroxylation, efficacy and side effects in Chinese patients with systemic lupus erythematosus. *Br J Clin Pharmacol*. 2016;81(2):327-340.

95. Hoglund J, Rafati N, Rask-Andersen M, et al. Improved power and precision with whole genome sequencing data in genome-wide association studies of inflammatory biomarkers. *Sci Rep*. 2019;9(1):16844.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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